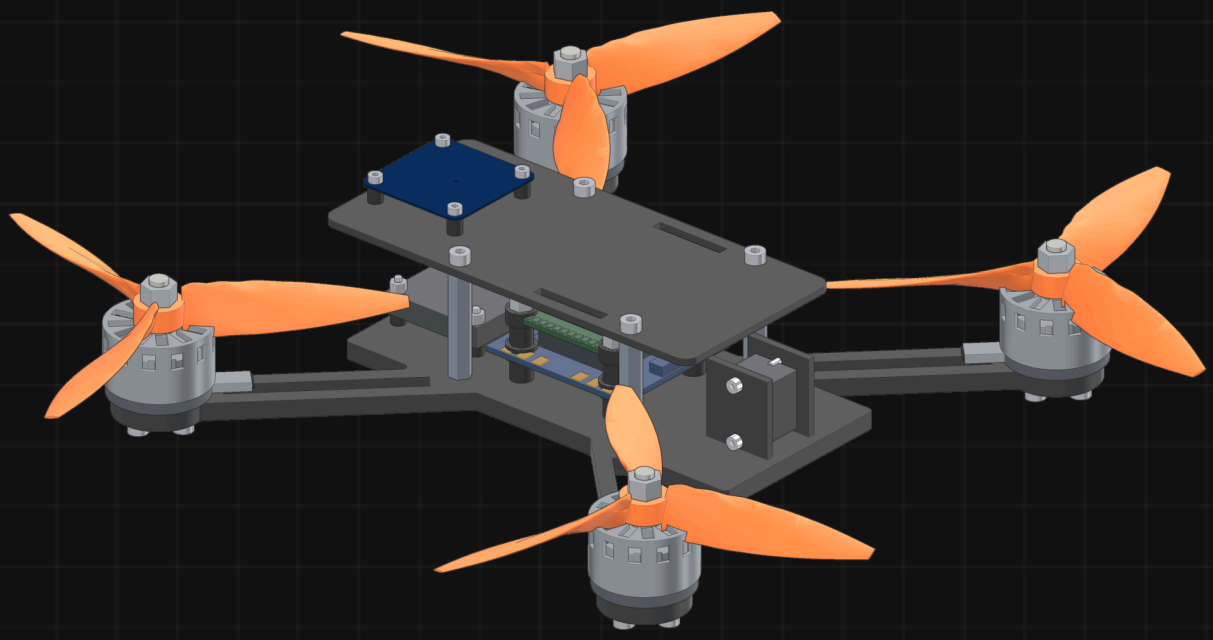
DATASHEET · REFERENCE SYSTEM

5-inch 6S Quadcopter

Reference 5-inch 6S freestyle quadcopter: DolphinRC F405 V3 + AM32 60A stack, four MEPS NEON 2207 V2 1950KV motors with HQProp Ethix S5 props, CNHL 1100 mAh 6S pack, RadioMaster RP1 V2 ELRS receiver, DJI O4 Air Unit, Matek M9N-5883 GNSS/compass, custom frame.



Assembled from UHD mates (16 placed bodies, 84 fasteners); strapped battery not shown.

STATIC THRUST	BATTERY	CAPACITY
6.3 kg[D]	**22.2** V	**1100** mAh
4 motors, full throttle, maker's test prop	6S LiPo	XT60
PEAK CURRENT	**ALL-UP WEIGHT**	**VIDEO LATENCY**
171.9 A	**587** g[A]	**20** ms
4 × motor 60 s peak	systemMass; cables and capacitor not included	DJI O4, Goggles 3

ABOUT THIS DOCUMENT

Every value is queried from the UHD model at build time and re-checked by `uhd-tech-docs-verify`. Markers show where a value is not a cited source value.

[A] Assumption: the model marks this value as an assumption (assumption trait)

[D] Derived: computed from model values; formula in the provenance appendix

[S] Stand-in: stand-in data derived from the model, not measured

[–] Not in model: the model does not state this value (data gap)

Unmarked values are cited from the part's sources (sources.json) or are design values of the model.

00 The product

WHAT IT IS

A 5-inch freestyle FPV quadcopter on 6S power, built from purchased modules on a custom true-X frame, with a digital video link, an ExpressLRS radio link, and GNSS with a compass.

FEATURES

- True-X 5-inch frame, custom, 225 mm wheelbase, with at least 5 mm of prop clearance
- 6S LiPo power: CNHL 1100 mAh 100C pack, DolphinRC F405 V3 + AM32 60A stack, MEPS NEON 2207 V2 1950KV motors, HQProp Ethix S5 props
- ExpressLRS radio link (RadioMaster RP1 V2)
- DJI O4 Air Unit digital video with the flight controller's OSD
- Matek M9N-5883 GNSS and compass
- Purchased modules and an existing flight-control stack: no PCB or flight-algorithm development

WHAT IT DOES

ID	FUNCTION	DONE BY
F1	Fly under radio control: the flight controller reads the pilot's sticks over the ExpressLRS link and drives the four motors through the ESC	RadioMaster RP1 V2, DolphinRC F405 V3, AM32 60A 4-in-1 ESC, MEPS NEON 2207 motors
F2	Send live digital video to the goggles, with the flight controller's OSD over it	DJI O4 Air Unit
F3	Report GNSS position and compass heading to the flight controller	Matek M9N-5883
F4	Power everything from one 6S LiPo pack	CNHL Black Series 1100 mAh 6S, XT60 lead, bulk capacitor

REQUIREMENTS

ID	REQUIREMENT	HOW IT IS SHOWN	PASS
R1	True-X frame, 225 mm wheelbase	Calipers or a rule between the shaft centres of diagonally opposite motors, both diagonals	225 mm on both diagonals, within 1 mm (target)
R2	≥ 5 mm prop clearance	Props fitted, motors unpowered: turn each prop by hand and measure the closest gap to the frame, the stack, the air unit, the antennas and the neighbouring prop	every gap ≥ 5 mm
R3	Power and propulsion as specified: CNHL 1100 mAh 6S 100C, DolphinRC F405 V3 + AM32 60A, MEPS NEON 2207 V2 1950KV, HQProp Ethix S5	Inspect each part's marking and its order line against system/bom.csv	every part number matches
R4	ELRS radio link (RadioMaster RP1 V2)	Transmitter bound; Betaflight Receiver tab; then switch the transmitter off	all channels follow the sticks; the failsafe flag shows within 2 s of switching off (target)
R5	DJI O4 digital video with OSD	Goggles linked to the air unit, quad on the bench supply	live image and Betaflight's OSD elements in the goggles
R6	GNSS position and compass heading (Matek M9N-5883)	Outdoors with open sky; Betaflight GPS and Sensors tabs; turn the quad through 360°	a 3D fix; the heading follows the nose (north within 15° of a hand compass, target)
R7	Purchased modules only; no PCB or flight-algorithm work	Inspect the build against system/bom.pdf	only bought, included and cut parts; stock Betaflight firmware
R8	Pack current within its rating	Blackbox log (or OSD peak current) over a flight with full-throttle punch-outs; the pack's temperature by hand or IR right after landing	peak current ≤ 220 A (the pack's burst rating); punch-outs shorter than a few seconds (docs/decisions.md D-Q05)
R9	Acceptance: model checks pass, frame cut from its CAD, assembled per the guide, measured data replaces the stand-ins	protoboard check; this package's work order and test record; docs/test-data/ status measured	0 errors; every check recorded; every test-data file measured

From docs/brief.md.

01 Overview

QUADCOPTER-5IN · V0.1.0

Reference 5-inch 6S freestyle quadcopter: DolphinRC F405 V3 + AM32 60A stack, four MEPS NEON 2207 V2 1950KV motors with HQProp Ethix S5 props, CNHL 1100 mAh 6S pack, RadioMaster RP1 V2 ELRS receiver, DJI O4 Air Unit, Matek M9N-5883 GNSS/compass, custom frame.

ROLE	MODULE	QTY
Frame	5-inch quadcopter frame (custom)	1
Top plate	5-inch frame top plate (custom)	1
FC + ESC stack	DolphinRC F405 V3 + AM32 60A stack	1
Arms	Arm (motor + propeller)	4
6S battery	CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)	1
ELRS receiver	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	1
DJI O4 Air Unit	DJI O4 Air Unit	1
GNSS + compass	Matek M9N-5883 GNSS & Compass	1
Battery strap	HGLRC Kevlar battery strap 20x250mm	1
Bulk capacitor 470 µF	Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor	1
Solder wire	Flux-cored solder wire, 0.8 mm	1
Flux	No-clean flux (pen or paste)	1
Threadlocker	Threadlocker, medium strength (blue)	1
Receiver heat-shrink sleeve	Clear heat-shrink sleeve for the receiver, about 15 mm flat width	1
Foam mounting tape	Double-sided foam mounting tape, 1 mm	1
Cable ties	Cable ties, 2.5 mm x 100 mm	1

CONTENTS

00	The product	02
01	Overview	04
02	Specifications	05
03	Mechanical	06
04	Interfaces and pinout	09
05	Electrical characteristics	10
06	Performance	11
07	Design checks	13
08	Bill of materials	14
09	Evidence and assumptions	17
10	Value provenance	20

SYSTEM ARCHITECTURE

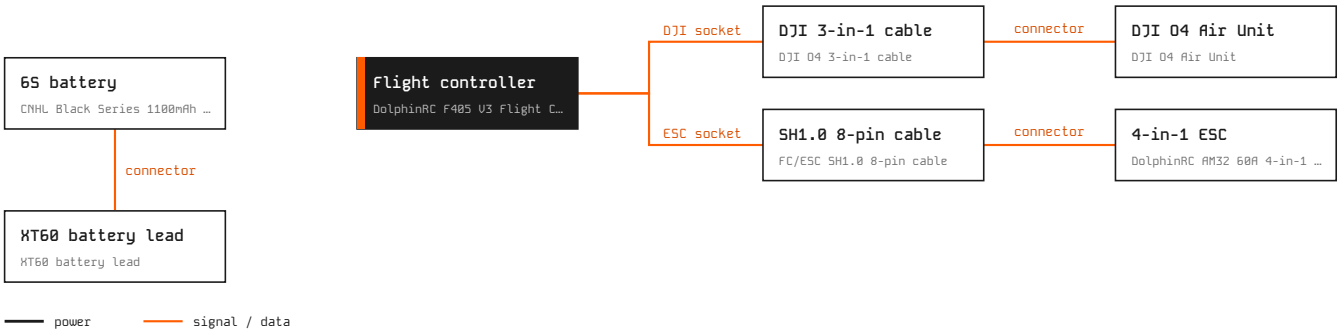


Figure 1 Power and signal links as stored in the model (validated with validateLinks); labels are the hub-side interfaces.

02 Specifications

PARAMETER	VALUE
AIRFRAME	
Layout	true-X, 5-inch
Wheelbase	225 mm
Bottom plate	5 mm
Standoff height	30 mm
Frame mass carbon, from CAD	86.9 g[A]
Stack pattern square, M3	30.5 mm
Motor pattern bolt circle, M3	16 mm
PROPULSION	
Motor	MEPS NEON 2207 V2 1950KV Brushless Motor
KV	1950 rpm/V
Stator / poles	12N14P
Peak current (60 s)	42.97 A
Max power	1036 W
Motor mass	36 g
Propeller	HQProp Ethix S5 5x4x3 Propeller
Prop mass	3.7 g
ESC per channel continuous	60 A
ESC burst	65 A
ESC firmware	AM32 (32-bit); firmware target AT32DEVF421
FLIGHT CONTROL	
Flight controller	DolphinRC F405 V3 Flight Controller
MCU	STM32F405RGT6, Cortex-M4 @ 168MHz
IMU	ICM-42688-P
Firmware target	Betaflight MATEKF405TE
UARTs	6
Receiver	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver
RC link	ExpressLRS
CRSF baud rate	420 000 Bd

PARAMETER	VALUE
VIDEO	
Air unit	DJI O4 Air Unit
Image sensor	1/2-inch CMOS
Field of view	117.6°
Max bitrate	100 Mbit/s
Supply range	3.7–13.2 V
Operating temperature	-10–40 °C
POWER	
Battery	CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)
Nominal voltage	22.2 V
Capacity	1100 mAh
Energy	24.42 Wh[D]
Continuous current	110 A
Burst current	220 A
Pack mass	211 g
All-up weight excl. cables, cap	587.2 g[A]
ESC input	6–30 V
BEC 5 V rating	3 A
BEC 10 V rating	2.5 A
GPS rail	4.5–5.0 V
NAVIGATION	
GNSS module	Matek M9N-5883 GNSS & Compass
Receiver	u-blox NEO-M9N (concurrent reception of GPS, Galileo, GLONASS, BeiDou)
Compass address	0x2C
Supply current	50 mA
GPS rail fed from assumed	bec_5v[A]

[A] Assumption: the model marks this value as an assumption (assumption trait)

[D] Derived: computed from model values; formula in the provenance appendix

[S] Stand-in: stand-in data derived from the model, not measured — Not in model: the model does not state this value (data gap)

Unmarked values are cited from the part's sources (sources.json) or are design values of the model.

03 Mechanical

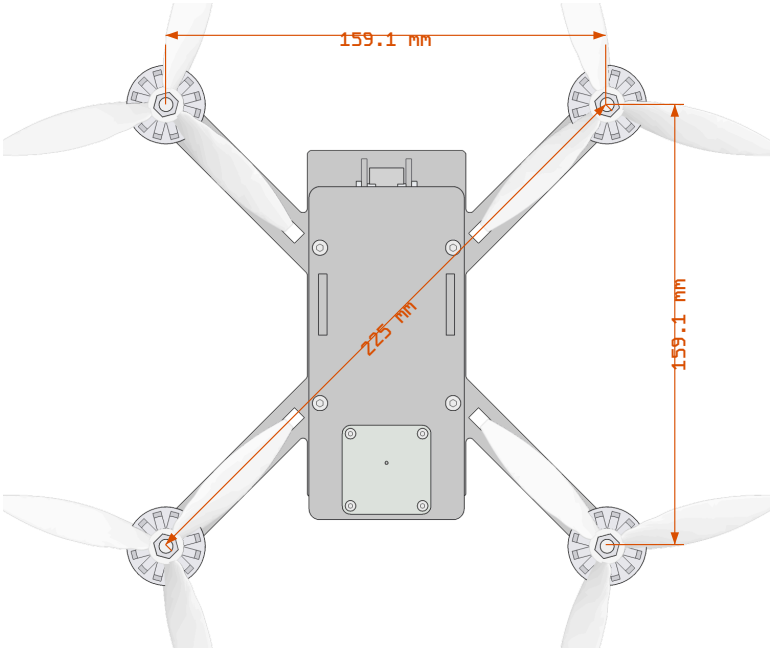


Figure 2 Top view, front up. Wheelbase is the frame's stated value; arm spacings are measured between the placed motor-mount frames. (unit: mm)

ENVELOPE

PARAMETER	VALUE
Wheelbase	225 mm
Arm spacing (x)	159.1 mm[D]
Arm spacing (y)	159.1 mm[D]
Plate thickness	5 mm
Standoff height	30 mm
Propeller diameter	5 in
Motor diameter	28.3 mm
Plate material	3K woven carbon fibre / epoxy laminate[A]
Material density	1.52 g/cm³[A]
Frame mass	86.9 g[A]
Top plate mass	19.8 g[A]

Plate masses are the generated CAD volume × the declared material density (systemMass), so they follow the design.

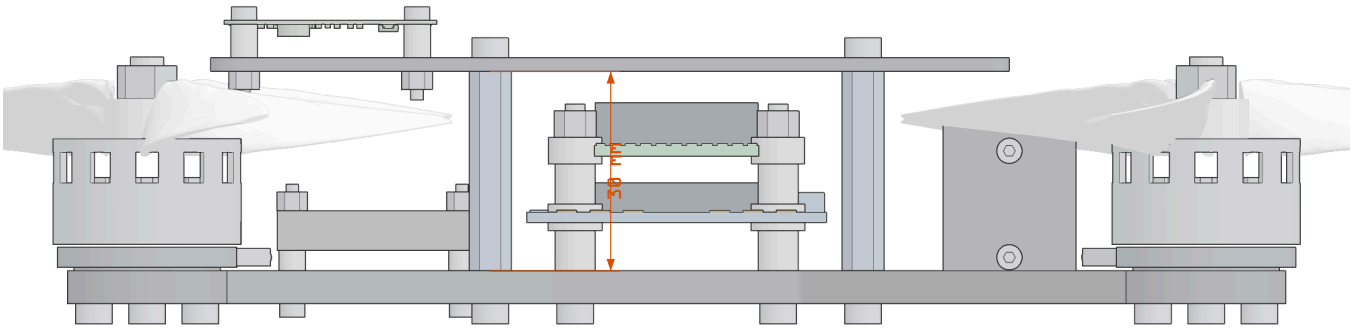


Figure 3 Right side, front to the right. Plate gap is the frame's standoff height. (unit: mm)

MASS BUDGET

PART	QTY	EACH	BASIS
CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)	1	211 g	stated
MEPS NEON 2207 V2 1950KV Brushless Motor	4	36 g	stated
5-inch quadcopter frame	1	86.9 g[A]	CAD × 3K woven carbon fibre / epoxy laminate
5-inch frame top plate	1	19.8 g[A]	CAD × 3K woven carbon fibre / epoxy laminate
HQProp Ethix S5 5x4x3 Propeller	4	3.7 g	stated
M3 × 8 mm motor screw	16	0.91 g[A]	CAD × A2 stainless steel
Matek M9N-5883 GNSS & Compass	1	14.5 g	stated
DolphinRC AM32 60A 4-in-1 ESC	1	13.2 g	stated
M3 × 30 mm socket head cap screw	4	2.3 g[A]	CAD × A2 stainless steel
HGLRC Kevlar battery strap 20x250mm	1	8.3 g	stated
DJI O4 Air Unit	1	8.2 g	stated
DolphinRC F405 V3 Flight Controller	1	7.7 g	stated
M3 × 8 mm socket head cap screw	8	0.91 g[A]	CAD × A2 stainless steel
M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF	4	1.82 g	stated
M5 flanged lock nut (prop nut, included with the motor)	4	1.4 g[A]	CAD × A2 stainless steel
M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	4	0.61 g[A]	CAD × A2 stainless steel
M2 × 18 mm socket head cap screw	4	0.57 g[A]	CAD × A2 stainless steel
RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	1	2.2 g	stated
M2 nyloc nut, DIN 985, A2 stainless	8	0.24 g[A]	CAD × A2 stainless steel
M2 × 12 mm socket head cap screw	4	0.43 g[A]	CAD × A2 stainless steel
ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25	8	0.15 g	stated
M2 × 5 mm socket head cap screw	4	0.27 g[A]	CAD × A2 stainless steel
M3 rubber grommet, ESC	4	0.2 g[A]	CAD × Silicone rubber
M3 rubber grommet, FC	4	0.19 g[A]	CAD × Silicone rubber
M2 nylon spacer, Ø2.7/4.0 × 3 mm	4	0.05 g	stated
M2 nylon spacer, Ø2.7/4.0 × 5 mm	4	0.05 g	stated

All-up weight 587.2 g[A] (systemMass), of which 154.4 g[A] is computed from assumed material or nominal CAD geometry. Not included, no mass in the model: FC/ESC SH1.0 8-pin cable ×1, Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor ×1, XT60 battery lead (12 AWG) ×1, DJI O4 3-in-1 cable ×1, GNSS lead (GH6P, FC end soldered) ×1, Receiver wires (RP1 V2 CRSF wire) ×1, Bulk capacitor leads ×1, Motor 2 phase leads (Front-right) ×1, Motor 1 phase leads (Rear-right) ×1, Motor 3 phase leads (Rear-left) ×1, Motor 4 phase leads (Front-left) ×1, Flux-cored solder wire, 0.8 mm ×1, No-clean flux (pen or paste) ×1, Threadlocker, medium strength (blue) ×1, Clear heat-shrink sleeve for the receiver, about 15 mm flat width ×1, Double-sided foam mounting tape, 1 mm ×1, Cable ties, 2.5 mm × 100 mm ×1. Rows in order of total mass (quantity × each).

INTERFACE GEOMETRY

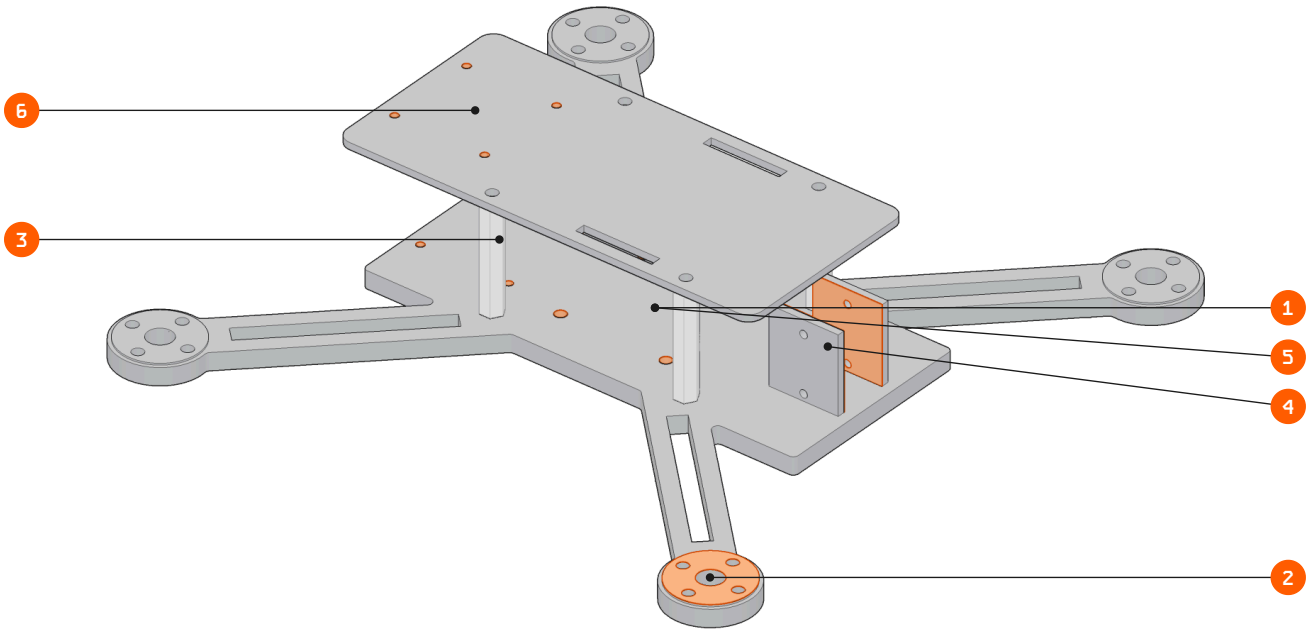


Figure 4 Structural interfaces of the frame and top plate (accent), from each interface's geometry frame and CAD feature binding.

#	INTERFACE	PATTERN	FASTENER	MATES WITH	BINDING
1	**Stack mount** frame.stack_mount	4 × on 30.5 mm square	3 mm	DolphinRC AM32 60A 4-in-1 ESC via stack_hardware DolphinRC F405 V3 Flight Controller via stack_hardware	D1 D2 D3
2	**Front-right motor mount** frame.motor_mount_fr	4 × on Ø16 mm circle	3 mm	MEPS NEON 2207 V2 1950KV Brushless Motor via arm_fr_hardware	D1 D2 D3
3	**Transmission module mount** frame.vtx_mount	4 × on 25.5 mm square	2 mm	DJI O4 Air Unit via vtx_hardware	D1 D2 D3
4	**Camera side plates** frame.camera_mount	4 × on 16 mm × 14 mm rectangle	2 mm	DJI O4 Air Unit via camera_hardware	D1 D2
5	**Top-plate standoffs** frame.standoff_mount	4 × on 56 mm × 48 mm rectangle	3 mm	5-inch frame top plate (custom) via frame_hardware	D1 D2 D3
6	**GPS mount** top_plate.gps_mount	4 × on 26 mm square	2 mm	Matek M9N-5883 GNSS & Compass via gps_hardware	D1 D2 D3

Binding: D1 named CAD feature, D2 per-interface artifact, D3 procedural from parameters (docs/geometry-artifacts.md).

04 Interfaces and pinout

Allocation of the DolphinRC F405 V3 Flight Controller (`stack/fc`) interfaces, pad to pad, from the stored links. Crossovers (TX→RX) are marked ×. The link state is the result of pairwise DRC.

HUB INTERFACE	PADS (HUB → PEER)	PEER	PROTOCOL	PARAMETERS	STATE
DJI socket (6-pin)	10V → P1 GND P2 → P2 TX5 → P3 R5 → P4 GND P5 → P5 SBUS → P6	DJI O4 3-in-1 cable FC end (DJI 6-pin)	CONNECTOR	—	CONFIGURED

05 Electrical characteristics

BEC and rail utilisation (known loads)



Figure 5 checkSystem supply_budget.

Propulsion peak current vs battery ratings

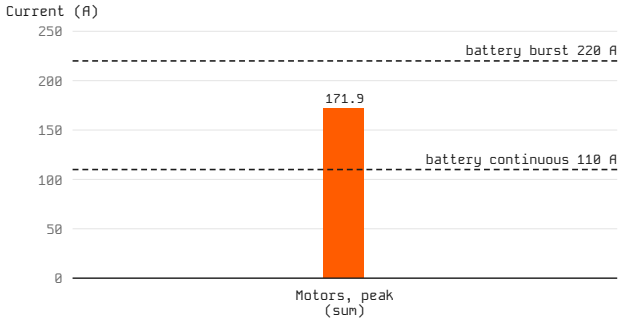


Figure 6 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.

POWER RAILS

RAIL	VOLTAGE	CAPACITY	KNOWN LOAD	LOADS	UNBUDGETED
5V BEC stack:bec_5v	5.0 V	15 W	0.225 W	receiver:vcc_5v, gnss:vin_5v, stack:rail_4v5	receiver:vcc_5v
10V/12V BEC stack:bec_10v	10.0 V	25 W	10 W	video:vcc	—
4.5V stack:rail_4v5	4.5–5.0 V	from stack:bec_5v A	in parent	stack:bec_5v, gnss:vin_5v	budgeted on stack:bec_5v (assumed branch)

From checkSystem rule supply_budget: capacity = rail voltage × current rating; loads that state no draw cannot be budgeted. A branch rail (supplied_from) is budgeted on the output that feeds it; A marks a relation assumed, not cited.

SUPPLY INPUTS

MODULE	INPUT	VOLTAGE RANGE	MIN. SUPPLY POWER
DolphinRC F405 V3 Flight Controller	VBAT VBAT	6.0–30.0 V	—
DolphinRC AM32 60A 4-in-1 ESC	BAT+ +	6.0–30.0 V	—
RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	5V 5V	5.0 V	—
DJI O4 Air Unit	VCC (3-in-1 cable, red) VCC	3.7–13.2 V	10 W
Matek M9N-5883 GNSS & Compass	5V 5V	5.0 V	—

Stored links: 48; configured: 48. Diagnostics on the nominal build: warning propulsion_current, info supply_budget, info supply_budget, info supply_budget, info prop_handedness, info fastener_torque.

06 Performance

TEST DATA

MOTOR THRUST TABLES (MANUFACTURER)

THROTTLE	SZ4942 V	A	RPM	G	W	SZ5145 V	A	RPM	G	W
10 %	25.2	0.3	4888	28	9	25.2	0.4	4854	33	9
20 %	25.2	1.3	9570	119	33	25.2	1.4	9465	143	35
30 %	25.1	3.3	14 157	277	83	25.1	3.6	13 911	334	90
40 %	25.1	5.8	17 710	452	145	25.0	6.4	17 304	533	159
50 %	25.0	9.5	20 929	648	237	24.9	10.4	20 457	763	260
60 %	24.9	14.3	23 893	865	355	24.8	15.5	23 263	1000	384
70 %	24.7	20.0	26 534	1083	494	24.7	21.5	25 704	1233	531
80 %	24.5	27.3	28 890	1298	668	24.5	28.7	27 751	1446	701
90 %	24.3	35.2	30 766	1470	855	24.3	36.8	29 199	1580	893
100 %	24.2	41.8	31 973	1584	1009	24.1	43.0	30 342	1699	1036

MEPS NEON 2207 V2 1950KV Brushless Motor: every row of the maker's KV table, per test propeller (MEPS SZ4942 at 25.2 V; MEPS SZ5145 at 25.2 V); columns per propeller are measured supply voltage, current, speed, thrust and electrical power. The fitted propeller (HQProp Ethix S5 5x4x3 Propeller) is not one the maker tested, so these figures, and every figure derived from them below, are for the maker's test props.

⚠ **CAUTION** Charts and figures tagged STAND-IN · NOT MEASURED are derived from the model to show the expected shape. Replace them by dropping measured files into `docs/test-data/` (status `measured`); the next build swaps them in.

Thrust and current per motor vs throttle

STAND-IN · NOT MEASURED

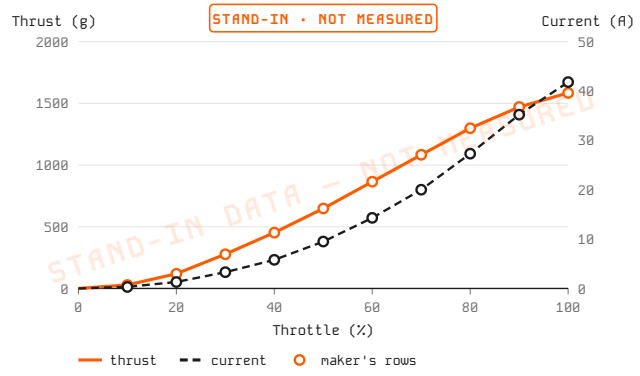


Figure 7 Circles: every row of the maker's table (MEPS SZ4942 at 25.2 V, not the fitted prop). Lines: Stand-in: the motor maker's thrust table row by row, scaled to all motors, plus avionics draw.

Flight time by regime STAND-IN · NOT MEASURED

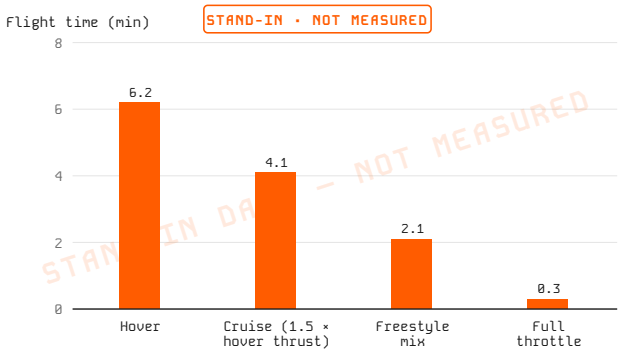


Figure 8 Stand-in: time = capacity × usable fraction ÷ average current. Static-thrust currents: pessimistic for forward flight.

Battery discharge STAND-IN · NOT MEASURED

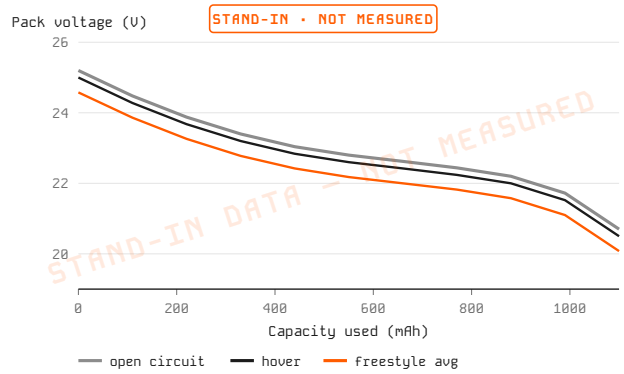


Figure 9 Stand-in: generic LiPo open-circuit curve per cell minus I·R sag, scaled to the pack.

ESC temperature over a punch-out STAND-IN · NOT MEASURED

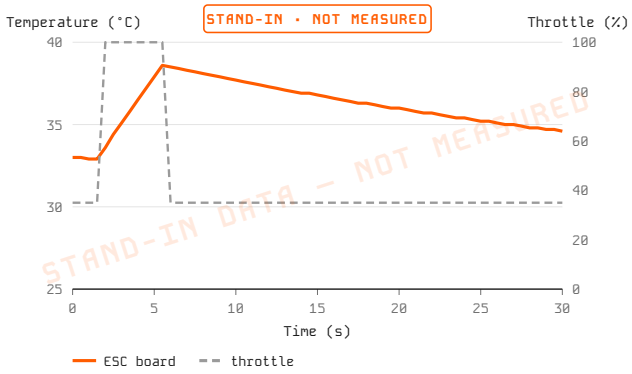


Figure 10 Stand-in: lumped thermal model C·dT/dt = P − (T − ambient)/R with conduction and switching losses per channel.

PERFORMANCE SUMMARY

FIGURE	VALUE	TEST DATA
All-up weight (estimate)	607 g[S]	hover-current
Hover current	8.5 A[S]	hover-current
Hover power	189 W[S]	hover-current
Avionics current	0.7 A[S]	hover-current
Total static thrust at 100 %	6.34 kg[S]	throttle-sweep
Total current at 100 %	167.9 A[S]	throttle-sweep
Thrust-to-weight at 25.2 V[S]	10.4[S] : 1	thrust-to-weight
Thrust-to-weight at 22.2 V[S]	8.1[S] : 1	thrust-to-weight
Thrust-to-weight at 21.0 V[S]	7.2[S] : 1	thrust-to-weight
Flight time, hover	6.2 min[S]	flight-time
Flight time, freestyle mix	2.1 min[S]	flight-time
ESC peak temperature	38.6 °C[S]	esc-temp-punchout

STAND-IN ASSUMPTIONS

- motor figures interpolate linearly between the 10 rows of the maker's table (0 at 0 % throttle); pack current = table electrical power ÷ 22.2 V nominal
- maker's table used the MEPS SZ4942 prop at 25.2 V, not the fitted propeller
- avionics: DJI O4 minimum supply power 10 W, GNSS 50 mA at 5 V, FC + receiver 2.5 W (assumed), BEC efficiency 85 % (assumed)
- all-up weight = 587.2 g from systemMass (derived:mass.all_up, of which 154.4 g is CAD volume × assumed material) + 20 g assumed for the 17 parts with no mass in the model (FC/ESC SH1.0 8-pin cable, Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor, XT60 battery lead (12 AWG), DJI O4 3-in-1 cable, GNSS lead (GH6P, FC end soldered), Receiver wires (RP1 V2 CRSF wire), Bulk capacitor leads, Motor 2 phase leads (Front-right), Motor 1 phase leads (Rear-right), Motor 3 phase leads (Rear-left), Motor 4 phase leads (Front-left), Flux-cored solder wire, 0.8 mm, No-clean flux (pen or paste), Threadlocker, medium strength (blue), Clear heat-shrink sleeve for the receiver, about 15 mm flat width, Double-sided foam mounting tape, 1 mm, Cable ties, 2.5 mm × 100 mm)
- usable capacity 80 % of rated
- thermal resistance 2 K/W and capacity 20 J/K (board in prop wash) — not in the model
- effective conduction resistance 4 m ohm per channel and 1.5 W switching loss per channel at full throttle — not in the model
- starting 8 °C above ambient after the previous flight segment
- generic LiPo open-circuit voltage curve
- internal resistance 4 m ohm per cell — not in the model

07 Design checks

CHECKSYSTEM

The nominal build and each fault scenario in `scenarios.ts` run through `checkSystem` (the system and each circuit board in it). A fault scenario is expected to raise at least one error or warning.

SCENARIO	DIAGNOSTICS	FINDINGS (ERRORS AND WARNINGS)
Nominal build nominal	0 ERR 1 WARN 5 INFO	PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
O4 wired to raw 6S o4-on-vbat	1 ERR 1 WARN 4 INFO	LINK_STATE O4 VCC on the pack: incompatible — voltage: 18–25.2 V vs 3.7–13.2 V (ranges do not overlap) PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
20 × 20 frame for a 30.5 stack frame-20mm-stack	2 ERR 1 WARN 5 INFO	LINK_STATE ESC on frame: incompatible — hole_spacing: 30.5 mm vs 20 mm (ranges do not overlap); fastener_diameter: 3 mm vs 2 mm (ranges do not overlap) LINK_STATE FC on frame: incompatible — hole_spacing: 30.5 mm vs 20 mm (ranges do not overlap); fastener_diameter: 3 mm vs 2 mm (ranges do not overlap) PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
GPS on the receiver's UART gps-on-rx-uart	1 ERR 1 WARN 5 INFO	INTERFACE_REUSE stack:uart2 is linked 2 times (rx_fc_pads+rx_pads~crsf~uart2, gnss_pads+gnss_socket~uart2~uart_gnss) but supports 1. PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
GNSS lead with RX and TX swapped gnss-rx-tx-swapped	1 ERR 1 WARN 5 INFO	LINK_STATE UART1 ↔ GNSS UART (NEO-M9N): incompatible — uart1_rx is wired to uart_gnss_rx, but uart pairs it with uart_gnss_tx; uart1_tx is wired to uart_gnss_tx, but uart pairs it with uart_gnss_rx PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
Two compasses on one I2C bus two-compasses	1 ERR 1 WARN 5 INFO	BUS_ADDRESS stack:i2c1: 2 devices share I2C address 0x2C (gnss_2:i2c_compass, gnss:i2c_compass). PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.
M2 × 10 on the GPS short-gps-screws	0 ERR 1 WARN 5 INFO	PROPULSION_CURRENT 4 motors can peak at 171.88 A together, above the battery continuous rating 110 A but within its burst rating 220 A. Sustained full throttle will exceed the pack rating.

08 Bill of materials

DOCS/QUADCOPTER-5IN-BOM.JSON

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
1	Cable ties, 2.5 mm × 100 mm quad-cable-ties	—–	1	PURCHASE	—	—– A 1 · gaps 1
2	Clear heat-shrink sleeve for the receiver, about 15 mm flat width quad-receiver-heat-shrink	—–	1	PURCHASE	—	—– A 1 · gaps 1
3	CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60) cnhl-black-series-1100mah-6s-100c	CNHL (China Hobby Line) 1101006BK	1	PURCHASE	211 g	audited 18/18 A 3 · gaps 7
4	DJI O4 Air Unit dji-o4-air-unit	DJI DF3L2904	1	PURCHASE	8.2 g	audited 32/34 A 1 · gaps 5
5	DolphinRC AM32 60A 4-in-1 ESC dolphinrc-am32-60a-4in1-esc	DolphinRC (Shenzhen DolphinRC Model Technology Co., Ltd.) DolphinRC AM32 ESC 60A (F405 V3 stack, SKU DPDZ00041 option '60A ESC')	1	PURCHASE	13.2 g	audited 18/18 A 1 · gaps 5
6	DolphinRC F405 V3 Flight Controller dolphinrc-f405-v3-flight-controller	DolphinRC (Shenzhen DolphinRC Model Technology Co., Ltd.) DolphinRC F405 V3 (F405 V3 stack, SKU DPDZ00041 option 'F405-FC')	1	PURCHASE	7.7 g	audited 28/30 A 1 · gaps 8
7	Double-sided foam mounting tape, 1 mm quad-foam-mount-tape	—–	1	PURCHASE	—	—– A 1 · gaps 1
8	ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 ettinger-005-83-060-m3-nylon-spacer-6mm	ETTINGER 005.83.060	8	PURCHASE	0.15 g	audited 12/12 A 1 · gaps 1
9	Flux-cored solder wire, 0.8 mm quad-solder-wire	—–	1	PURCHASE	—	—– A 1 · gaps 1
10	HGLRC Kevlar battery strap 20x250mm hglrc-kevlar-battery-strap-20x250mm	HGLRC F.2D.02.00010	1	PURCHASE	8.3 g	audited 12/12 A 1 · gaps 5
11	HQProp Ethix S5 5x4x3 Propeller hqprop-ethix-s5	HQProp EthixS5-GR-PC	4 pack of 4	PURCHASE	3.7 g	audited 14/14 A 0 · gaps 4
12	M2 nyloc nut, DIN 985, A2 stainless iso-10511-m2-nyloc-nut	Westfield Fasteners WF16826	8	PURCHASE	0.24 g[A]	audited 10/10 A 1 · gaps 2
13	M2 nylon spacer, Ø2.7/4.0 × 3 mm m2-nylon-spacer-3mm	Ettinger 007.02.403	4 pack of 100	PURCHASE	0.05 g	audited 11/11 A 1 · gaps 1
14	M2 nylon spacer, Ø2.7/4.0 × 5 mm m2-nylon-spacer-5mm	Ettinger 007.02.405	4 pack of 100	PURCHASE	0.05 g	audited 11/11 A 1 · gaps 1
15	M2 × 12 mm socket head cap screw, ISO 4762, A2 stainless iso-4762-m2x12	Generic (ISO 4762 / DIN 912) WF2332	4	PURCHASE	0.43 g[A]	audited 11/11 A 1 · gaps 1
16	M2 × 18 mm socket head cap screw, ISO 4762, A2 stainless iso-4762-m2x18	Generic (ISO 4762 / DIN 912) WF17972	4	PURCHASE	0.57 g[A]	audited 11/11 A 1 · gaps 1
17	M2 × 5 mm socket head cap screw, ISO 4762, A2 stainless iso-4762-m2x5	Generic (ISO 4762 / DIN 912) WF2328	4	PURCHASE	0.27 g[A]	audited 14/14 A 1 · gaps 1

Rows as written by the document build (docs/quadcopter-5in-bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

BILL OF MATERIALS (CONTINUED)

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
18	M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF m3-aluminium-standoff-30mm	Ettinger 005.03.308	4 pack of 100	PURCHASE	1.82 g	audited 12/12 A 1 · gaps 1
19	M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless iso-10511-m3-nyloc-nut	Generic (ISO 10511 / DIN 985) WF1288	4	PURCHASE	0.61 g[A]	audited 12/12 A 1 · gaps 2
20	M3 × 30 mm socket head cap screw, ISO 4762, A2 stainless iso-4762-m3x30-socket-head-cap-screw	Generic (ISO 4762 / DIN 912) WF2352	4	PURCHASE	2.3 g[A]	audited 12/12 A 1 · gaps 2
21	M3 × 8 mm socket head cap screw, ISO 4762, A2 stainless iso-4762-m3x8	Generic (ISO 4762 / DIN 912) WF2344	8	PURCHASE	0.91 g[A]	audited 11/11 A 1 · gaps 1
22	Matek M9N-5883 GNSS & Compass matek-m9n-5883	Matek Systems M9N-5883	1	PURCHASE	14.5 g	audited 26/26 A 2 · gaps 8
23	MEPS NEON 2207 V2 1950KV Brushless Motor meps-neon-2207-v2-1950kv	MEPS NEON 2207 V2 1950KV	4	PURCHASE	36 g	audited 20/20 A 0 · gaps 7
24	No-clean flux (pen or paste) quad-flux	—-	1	PURCHASE	—	—- A 1 · gaps 1
25	Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor panasonic-eeu-fr1v471	Panasonic Industry EEU-FR1V471	1	PURCHASE	—	audited 17/18 A 2 · gaps 3
26	RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver radiomaster-rp1-v2-elrs-2g4	RadioMaster RP1 V2	1	PURCHASE	2.2 g	audited 21/21 A 0 · gaps 4
27	Threadlocker, medium strength (blue) quad-threadlocker	—-	1	PURCHASE	—	—- A 1 · gaps 1
28	5-inch frame top plate (custom) quadcopter-5in-top-plate	Delta Robotics (custom) QUAD5-TOP	1	FABRICATE	19.8 g[A]	—- A 1 · gaps 0
29	5-inch quadcopter frame (custom) quadcopter-5in-frame	Delta Robotics (custom) QUAD5-FRAME	1	FABRICATE	86.9 g[A]	—- A 1 · gaps 0
30	Bulk capacitor leads panasonic-eeu-fr1v471-leads	Panasonic Industry	1	INCL. panasonic-eeu-fr1v471	—	—- A 0 · gaps 0
31	DJI O4 3-in-1 cable dji-o4-3in1-cable	DJI	1	INCL. dji-o4-air-unit	—	—- A 1 · gaps 0
32	FC/ESC SH1.0 8-pin cable dolphinrc-sh8-fc-esc-cable	DolphinRC	1	INCL. dolphinrc-f405-v3-flight-controller	—	—- A 1 · gaps 0
33	GNSS lead (GH6P, FC end soldered) matek-gh6p-gnss-lead	Matek Systems	1	INCL. matek-m9n-5883	—	—- A 3 · gaps 0
34	M3 rubber grommet, ESC (included with the stack) dolphinrc-m3-grommet-esc	DolphinRC M3*8.1mm Rubber (for ESC)	4	INCL. dolphinrc-am32-60a-4in1-esc	0.2 g[A]	audited 11/11 A 1 · gaps 2
35	M3 rubber grommet, FC (included with the stack) dolphinrc-m3-grommet-fc	DolphinRC M3*8mm Rubber (for FC)	4	INCL. dolphinrc-f405-v3-flight-controller	0.19 g[A]	audited 11/11 A 1 · gaps 2
36	M3 × 8 mm motor screw (included with the motor) motor-screw-m3x8	Included (unbranded) ISO 4762 M3x8	16	INCL. meps-neon-2207-v2-1950kv	0.91 g[A]	audited 10/10 A 1 · gaps 2

Rows as written by the document build (docs/quadcopter-5in-bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

BILL OF MATERIALS (CONTINUED)

#	PART	MANUFACTURER / P/N	QTY	SUPPLY	MASS	VERIFICATION
37	M5 flanged lock nut (prop nut, included with the motor) meps-m5-flanged-lock-nut	Included (unbranded) MEPS NEON 2207 V2 prop nut	4	INCL. meps-neon-2207-v2-1950kv	1.4 g[A]	audited 11/11 A 1 · gaps 2
38	Motor 1 phase leads (Rear-right) meps-phase-leads-m1	MEPS	1	INCL. meps-neon-2207-v2-1950kv	—	— A 2 · gaps 0
39	Motor 2 phase leads (Front-right) meps-phase-leads-m2	MEPS	1	INCL. meps-neon-2207-v2-1950kv	—	— A 2 · gaps 0
40	Motor 3 phase leads (Rear-left) meps-phase-leads-m3	MEPS	1	INCL. meps-neon-2207-v2-1950kv	—	— A 2 · gaps 0
41	Motor 4 phase leads (Front-left) meps-phase-leads-m4	MEPS	1	INCL. meps-neon-2207-v2-1950kv	—	— A 2 · gaps 0
42	Receiver wires (RP1 V2 CRSF wire) radiomaster-rp1-crsf-wire	RadioMaster	1	INCL. radiomaster-rp1-v2-elrs-2g4	—	— A 2 · gaps 0
43	XT60 battery lead (12 AWG) dolphinrc-xt60-battery-lead	DolphinRC	1	INCL. dolphinrc-am32-60a-4in1-esc	—	— A 3 · gaps 0

Rows as written by the document build (docs/quadcopter-5in-bom.json). Mass per piece: the part's stated weight, else CAD volume × declared material density (derived); — = no mass in the model. A = assumption traits.

09 Evidence and assumptions

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
Cable ties, 2.5 mm x 100 mm	—	not audited	A specification: nylon 6.6 cable ties, 2.5 mm wide, 100 mm long	product; supplier
Clear heat-shrink sleeve for the receiver, about 15 mm flat width	—	not audited	A specification: clear polyolefin heat shrink, 2:1, flat width about 15 mm (fits the RP1's 10 x 10 mm board	product; supplier
CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)	5	18/18	—	minimum (discharge cut-off) cell voltage; operating / charging / storage temperature; burst duration; main lead and balance lead length; XT60 gender (+2)
DJI O4 Air Unit	5	32/34	—	antenna connector family; FC-side 3-in-1 cable connector family; power consumption / max input current; storage temperature; lens mount in box
DolphinRC AM32 60A 4-in-1 ESC	3	18/18	—	pin 1 orientation of 8-pin SH connector (silkscreen order only); burst current duration (5 s vs 10 s) (sources conflict); operating temperature (not stated); VBAT passthrough current limit on connector V pin (not stated); current sensor output voltage range (only Betaflight scale/offset given)
DolphinRC F405 V3 Flight Controller	4	28/30	A 4.5V rail supply: branch of bec_5v (5 V 3 A) via a diode	pin 1 orientation of all SH connectors (Manual shows silkscreen order only.); which UART the SBUS pad / DJI Sbus pin feeds, and inversion (Not stated; Betaflight target defaults serial RX to UART2.); 'ppm' silkscreen near RX3 pad (Low-res photo; target puts PPM on UART2 RX (PA3). Not claimed.); S1/S2 pad function (Product page says 10x PWM outputs (8 DShot), target defines SERVO1-3; S1/S2 modelled as PWM outputs only.); RSSI/Airspeed ADC pads (Listed as ADC channels but no pad labelled RSSI/Airspeed in drawings; omitted.) (+3)
Double-sided foam mounting tape, 1 mm	—	not audited	A specification: 1 mm closed-cell acrylic foam tape, cut to 10 x 10 mm and 10 x 16 mm	product; supplier
ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25	3	12/12	A stack-up inputs: bottom plate 5.0 mm carbon; grommeted board clamp height 4.0 mm each (ESC and FC)	operating temperature
Flux-cored solder wire, 0.8 mm	—	not audited	A specification: Sn63/Pb37 or a lead-free SAC alloy, rosin or no-clean flux core, 0.6–1.0 mm	product; supplier
HGLRC Kevlar battery strap 20x250mm	3	12/12	—	strap thickness; tensile strength / load rating; buckle dimensions and anti-slip strip length; temperature range; frame top-plate thickness and slot width
HQProp Ethix S5 5x4x3 Propeller	2	14/14	—	hub mounting style / retention; recommended motor / cell count; operating temperature; thrust / efficiency data
M2 nyloc nut, DIN 985, A2 stainless	2	10/10	A part choice: M2 nyloc nut, DIN 985, A2 stainless	height 2.8 mm, 4 mm across flats (DIN 985 Table 1 starts at M3 and the listing gives no M2 dimensions (data_gap)); manufacturer CAD (none published; nominal body generated)

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

EVIDENCE (CONTINUED)

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
M2 nylon spacer, Ø2.7/4.0 × 3 mm	2	11/11	A part choice: M2 nylon spacer, Ø2.7/4.0 × 3 mm	manufacturer CAD (none published; nominal body generated)
M2 nylon spacer, Ø2.7/4.0 × 5 mm	2	11/11	A part choice: M2 nylon spacer, Ø2.7/4.0 × 5 mm	manufacturer CAD (none published; nominal body generated)
M2 × 12 mm socket head cap screw, ISO 4762, A2 stainless	1	11/11	A length: 12 mm	manufacturer CAD (none published; nominal body generated)
M2 × 18 mm socket head cap screw, ISO 4762, A2 stainless	1	11/11	A length: 18 mm	manufacturer CAD (none published; nominal body generated)
M2 × 5 mm socket head cap screw, ISO 4762, A2 stainless	2	14/14	A length: 5 mm	manufacturer CAD (none published; nominal body generated)
M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF	2	12/12	A part choice: M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF	manufacturer CAD (none published; nominal body generated)
M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	2	12/12	A stack-up inputs: bottom plate 5.0 mm carbon; grommeted board clamp height 4.0 mm each (ESC and FC)	mass; nylon insert temperature limit
M3 × 30 mm socket head cap screw, ISO 4762, A2 stainless	2	12/12	A stack-up inputs: bottom plate 5.0 mm carbon; grommeted board clamp height 4.0 mm each (ESC and FC)	mass; operating temperature
M3 × 8 mm socket head cap screw, ISO 4762, A2 stainless	1	11/11	A length: 8 mm	manufacturer CAD (none published; nominal body generated)
Matek M9N-5883 GNSS & Compass	13	26/26	–	pin 1 of JST-GH-6P; edge pads VBus, D-, D+, Bt and V-USB jumper; fastener size for 2 mm holes; hole pattern Y spacing; logic levels, I2C pull-ups on module (+3)
MEPS NEON 2207 V2 1950KV Brushless Motor	5	20/20	–	lead wire gauge (AWG); continuous current rating; operating / storage temperature; maximum voltage (V); prop nut thread handedness / shaft rotation (+2)
No-clean flux (pen or paste)	–	not audited	A specification: no-clean rosin or synthetic flux, pen or syringe	product; supplier
Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor	3	17/18	A identity: Panasonic EEU-FR1V471 (ø10 × 16 mm) A lead polarity identification: negative lead marked by the sleeve stripe	bundled capacitor brand/series/MPN; mass; ESR (as such)
RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver	8	21/21	–	supply current; operating temperature; logic level of TX/RX pads; receiver sensitivity (RP1-specific)
Threadlocker, medium strength (blue)	–	not audited	A specification: medium-strength anaerobic threadlocker, removable with hand tools	product; supplier
5-inch frame top plate (custom)	–	not audited	A material: 3K woven carbon fibre, 1.52 g/cm³	–
5-inch quadcopter frame (custom)	–	not audited	A material: 3K woven carbon fibre, 1.52 g/cm³	–
Bulk capacitor leads	–	not audited	–	–

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

EVIDENCE (CONTINUED)

PART	SOURCES	AUDIT	ASSUMPTIONS	OPEN DATA GAPS
DJI O4 3-in-1 cable	—	not audited	A conductor gauge: 0.8 mm OD (about 30 AWG)	—
FC/ESC SH1.0 8-pin cable	—	not audited	A conductor gauge and colour: 0.95 mm OD, black	—
GNSS lead (GH6P, FC end soldered)	—	not audited	A installed length: 97 mm A colour per pin: red 5V, yellow RX, blue TX, green CL, white DA, black G A conductor gauge: 1.1 mm OD (about 26 AWG)	—
M3 rubber grommet, ESC (included with the stack)	1	11/11	A part choice: M3 rubber grommet, ESC (included with the stack)	grommet height 4 mm and outer Ø8.1 mm (only 'M3*8.1mm' is stated (data_gap)); manufacturer CAD (none published; nominal body generated)
M3 rubber grommet, FC (included with the stack)	1	11/11	A part choice: M3 rubber grommet, FC (included with the stack)	grommet height 4 mm and outer Ø8 mm (only 'M3*8mm' is stated (data_gap)); manufacturer CAD (none published; nominal body generated)
M3 × 8 mm motor screw (included with the motor)	3	10/10	A length: 8 mm	screw length 8 mm (not stated; chosen for the stack-up (assumption trait)); manufacturer CAD (none published; nominal body generated)
M5 flanged lock nut (prop nut, included with the motor)	3	11/11	A part choice: M5 flanged lock nut (prop nut, included with the motor)	this nut's own height, across flats, flange and thread hand (not stated by MEPS (data_gap)); manufacturer CAD (none published; nominal body generated)
Motor 1 phase leads (Rear-right)	—	not audited	A wire gauge: 20 AWG A installed length: 78 mm	—
Motor 2 phase leads (Front-right)	—	not audited	A wire gauge: 20 AWG A installed length: 78 mm	—
Motor 3 phase leads (Rear-left)	—	not audited	A wire gauge: 20 AWG A installed length: 78 mm	—
Motor 4 phase leads (Front-left)	—	not audited	A wire gauge: 20 AWG A installed length: 78 mm	—
Receiver wires (RP1 V2 CRSF wire)	—	not audited	A colour per pad: green TX, white RX, red 5V, black G A gauge and length: 1.1 mm OD, 41 mm	—
XT60 battery lead (12 AWG)	—	not audited	A lay: flat pair, not twisted A installed length: 30 mm A XT60 housing colour: yellow (AMASS standard)	—

Sources: sources.json entries. Audit: independent re-check of facts against the sources (verification.json).

10 Value provenance

This document shows 652 bound values. Each is a query into the UHD model, re-resolved and compared by `uhd-tech-docs-verify`; the full list with sources is in `uhd-quad5-ds.values.json` next to the document.

STATUS	VALUES	MEANING
Cited source	165	stated by a source in the part's sources.json
Model design value	401	a design value of a custom module, or a UHD check result
Derived	26	computed from model values; formula in the provenance appendix
Assumption	21	the model marks this value as an assumption (assumption trait)
Stand-in	15	stand-in data derived from the model, not measured
Not in model	24	the model does not state this value (data gap)

STATUS	VALUE	QUERY	DERIVATION / REASON
D Derived	6.3 kg	derived:propulsion.thrust_full_total	motors × thrust_g of the top (100 %) row of the first thrust table · manufacturer table with MEPS SZ4942 at 25.2 V (not the fitted propeller) · inputs: 1
A Assumption	587 g	derived:mass.all_up	systemMass: ∑ quantity × (weight_g, else CAD volume × material density) over every instance; instances with neither are listed, not guessed · excludes 17 part(s) with no mass: FC/ESC SH1.0 8-pin cable, Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic cap
A Assumption	86.9 g	derived:mass.module(quadcopter-5in-frame)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 57.15 cm³ × 1.52 g/cm³ (3K woven carbon fibre / epoxy laminate); assumption: DragonPlate's woven quasi-isotropic row is taken for 3K woven plate; the t
D Derived	24.42 Wh	derived:power.battery_energy_Wh	battery voltage × capacity · inputs: 2
A Assumption	bec_5v	def:dolphinrc-f405-v3-flight-controller:interfaces[id=rail_4v5].traits[type=supplied_from].params.interfaceId	assumption trait · https://dolphinrc.com/u_file/2608/24/file/DolphinRCF40520250616153927-165929bb5a.pdf
D Derived	159.1 mm	derived:frameDistance(frame.motor_mount_fl,frame.motor_mount_fr)	|placement(a) · a.frame.origin - placement(b) · b.frame.origin| from assemble() · inputs: 2
D Derived	159.1 mm	derived:frameDistance(frame.motor_mount_rr,frame.motor_mount_fr)	|placement(a) · a.frame.origin - placement(b) · b.frame.origin| from assemble() · inputs: 2
D Derived	159.1 mm	derived:frameDistance(frame.motor_mount_fl,frame.motor_mount_rl)	|placement(a) · a.frame.origin - placement(b) · b.frame.origin| from assemble() · inputs: 2
A Assumption	3K woven carbon fibre / epoxy laminate	def:quadcopter-5in-frame:domains[domain=mechanical].material.name	assumption trait · https://dragonplate.com/carbon-fiber-sheets
A Assumption	1.52 g/cm³	def:quadcopter-5in-frame:domains[domain=mechanical].material.density_g_cm3	assumption trait · https://dragonplate.com/carbon-fiber-sheets
A Assumption	19.8 g	derived:mass.module(quadcopter-5in-top-plate)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 13.05 cm³ × 1.52 g/cm³ (3K woven carbon fibre / epoxy laminate); assumption: DragonPlate's woven quasi-isotropic row is taken for 3K woven plate; the t
A Assumption	0.91 g	derived:mass.module(motor-screw-m3x8)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.11 cm³ × 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: The maker does not state the material; steel (A2 density) and the generator's nomin

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
A Assumption	2.3 g	derived:mass.module(iso-4762-m3x30-socket-head-cap-screw)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.28 cm³ × 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: The CAD body is library/cad/py/parts.py's nominal ISO 4762 shape (plain shank, no t
A Assumption	0.91 g	derived:mass.module(iso-4762-m3x8)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.11 cm³ × 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: Volume is the generator's nominal shape (plain shank, no thread form), not a measur
A Assumption	1.4 g	derived:mass.module(meps-m5-flanged-lock-nut)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.18 cm³ × 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: Nominal hex nut shape with the nylon insert counted as steel (a slight over-estimat
A Assumption	0.61 g	derived:mass.module(iso-10511-m3-nyloc-nut)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.08 cm³ × 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: The CAD body is library/cad/py/parts.py's nominal DIN 985 hex with the nylon insert
A Assumption	0.57 g	derived:mass.module(iso-4762-m2x18)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.07 cm³ × 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: Volume is the generator's nominal shape (plain shank, no thread form), not a measur
A Assumption	0.24 g	derived:mass.module(iso-10511-m2-nyloc-nut)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.03 cm³ × 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: Nominal hex nut shape with the nylon insert counted as steel (a slight over-estimat
A Assumption	0.43 g	derived:mass.module(iso-4762-m2x12)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.05 cm³ × 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: Volume is the generator's nominal shape (plain shank, no thread form), not a measur
A Assumption	0.27 g	derived:mass.module(iso-4762-m2x5)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.03 cm³ × 8 g/cm³ (A2 (304 / 1.4301) stainless steel); assumption: Volume is the generator's nominal shape (plain shank, no thread form), not a measur
A Assumption	0.2 g	derived:mass.module(dolphinrc-m3-grommet-esc)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.17 cm³ × 1.15 g/cm³ (Silicone rubber (assumed)); assumption: DolphinRC calls the grommets 'Rubber' and states no compound, size or weight; 1.15 g/cm³
A Assumption	0.19 g	derived:mass.module(dolphinrc-m3-grommet-fc)	weight_g if stated, else CAD body volume (generator manifest) × domains[mechanical].material.density_g_cm3 · 0.16 cm³ × 1.15 g/cm³ (Silicone rubber (assumed)); assumption: DolphinRC calls the grommets 'Rubber' and states no compound, size or weight; 1.15 g/cm³
A Assumption	154.4 g	derived:mass.assumed	Σ systemMass entries whose material carries an assumption · inputs: 13
S Stand-in	607 g	test:hover-current:summary[id=auw_g].value	docs/test-data/hover-current.json

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
S Stand-in	8.5 A	test:hover-current:summary[id=hover_current_A].value	docs/test-data/hover-current.json
S Stand-in	189 W	test:hover-current:summary[id=hover_power_W].value	docs/test-data/hover-current.json
S Stand-in	0.7 A	test:hover-current:summary[id=avionics_current_A].value	docs/test-data/hover-current.json
S Stand-in	6.34 kg	test:throttle-sweep:summary[id=thrust_total_full].value	docs/test-data/throttle-sweep.json
S Stand-in	167.9 A	test:throttle-sweep:summary[id=current_total_full].value	docs/test-data/throttle-sweep.json
S Stand-in	25.2 V	test:thrust-to-weight:rows[0][0]	docs/test-data/thrust-to-weight.json
S Stand-in	10.4	test:thrust-to-weight:rows[0][2]	docs/test-data/thrust-to-weight.json
S Stand-in	22.2 V	test:thrust-to-weight:rows[1][0]	docs/test-data/thrust-to-weight.json
S Stand-in	8.1	test:thrust-to-weight:rows[1][2]	docs/test-data/thrust-to-weight.json
S Stand-in	21.0 V	test:thrust-to-weight:rows[2][0]	docs/test-data/thrust-to-weight.json
S Stand-in	7.2	test:thrust-to-weight:rows[2][2]	docs/test-data/thrust-to-weight.json
S Stand-in	6.2 min	test:flight-time:summary[id=hover_min].value	docs/test-data/flight-time.json
S Stand-in	2.1 min	test:flight-time:summary[id=mix_min].value	docs/test-data/flight-time.json
S Stand-in	38.6 °C	test:esc-temp-punchout:summary[id=peak_C].value	docs/test-data/esc-temp-punchout.json
D Derived	0	derived:diagCount(nominal,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(nominal,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	5	derived:diagCount(nominal,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(o4-on-vbat,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(o4-on-vbat,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	4	derived:diagCount(o4-on-vbat,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	2	derived:diagCount(frame-20mm-stack,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(frame-20mm-stack,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	5	derived:diagCount(frame-20mm-stack,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(gps-on-rx-uart,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(gps-on-rx-uart,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	5	derived:diagCount(gps-on-rx-uart,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(gnss-rx-tx-swapped,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(gnss-rx-tx-swapped,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	5	derived:diagCount(gnss-rx-tx-swapped,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(two-compasses,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(two-compasses,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	5	derived:diagCount(two-compasses,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1

FLAGGED VALUES (CONTINUED)

STATUS	VALUE	QUERY	DERIVATION / REASON
D Derived	0	derived:diagCount(short-gps-screws,error)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	1	derived:diagCount(short-gps-screws,warning)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
D Derived	5	derived:diagCount(short-gps-screws,info)	count of checkSystem(scenario).diagnostics, plus each board below it, with the severity · inputs: 1
– Not in model	–	sys:bom[line=1].manufacturer	UHD bom
– Not in model	–	sys:bom[line=1].verification	UHD bom
– Not in model	–	sys:bom[line=2].manufacturer	UHD bom
– Not in model	–	sys:bom[line=2].verification	UHD bom
– Not in model	–	sys:bom[line=7].manufacturer	UHD bom
– Not in model	–	sys:bom[line=7].verification	UHD bom
– Not in model	–	sys:bom[line=9].manufacturer	UHD bom
– Not in model	–	sys:bom[line=9].verification	UHD bom
– Not in model	–	sys:bom[line=24].manufacturer	UHD bom
– Not in model	–	sys:bom[line=24].verification	UHD bom
– Not in model	–	sys:bom[line=27].manufacturer	UHD bom
– Not in model	–	sys:bom[line=27].verification	UHD bom
– Not in model	–	sys:bom[line=28].verification	UHD bom
– Not in model	–	sys:bom[line=29].verification	UHD bom
– Not in model	–	sys:bom[line=30].verification	UHD bom
– Not in model	–	sys:bom[line=31].verification	UHD bom
– Not in model	–	sys:bom[line=32].verification	UHD bom
– Not in model	–	sys:bom[line=33].verification	UHD bom
– Not in model	–	sys:bom[line=38].verification	UHD bom
– Not in model	–	sys:bom[line=39].verification	UHD bom
– Not in model	–	sys:bom[line=40].verification	UHD bom
– Not in model	–	sys:bom[line=41].verification	UHD bom
– Not in model	–	sys:bom[line=42].verification	UHD bom
– Not in model	–	sys:bom[line=43].verification	UHD bom